ANALYSIS

SOFTWARE ENGINEERING II

TESTIGOS DE MELUK

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Saturday, October 17th



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2026-2

Requirements

Functional Requirements

Module 1: Information Security

FR-01-01: The system must require every user to log in by entering their unique credentials (registered email and alphanumeric password). Upon successful validation of these credentials in the database, the system will not grant direct access to modules nor issue final access tokens (JWT). Instead, it will generate a temporary pre-authentication token valid solely to proceed to the two-factor verification phase (2FA). If the credentials entered are incorrect or the user does not exist, the system must deny access and display the error message "Correo electrónico o contraseña incorrectos".

FR-01-02: Following the successful validation of login credentials, and exclusively when the user does not have a previously configured authenticator app, the system must allow the initial binding of an external TOTP-based application. To complete this process, the system will generate and display a unique QR code along with its alphanumeric secret key in plain text, both valid for a maximum of 5 minutes. To complete and successfully confirm this pairing, the system will require the user to enter a valid code generated by the external application before the time limit expires, thereby enabling mandatory two-factor authentication for future logins. If the 5-minute validity window expires without completing the registration, the system will invalidate both the QR code and the alphanumeric secret key, display the on-screen message "El tiempo de vinculación ha expirado. Por favor, genere un nuevo código", and require the user to explicitly request the generation of a new code to restart the pairing process.

FR-01-03: Following the successful validation of login credentials, the system must require and validate a dynamic 6-digit temporary code based on the TOTP standard generated by the user's authenticator app. If the code entered is correct and currently valid, the system will complete the process by issuing the final access and refresh tokens under the JWT standard. Otherwise, the system will deny access and display the message "Código de verificación incorrecto o expirado".

FR-01-04: The system must restrict access to its modules, views, and operations by implementing a strict Role-Based Access Control (RBAC) model with three predefined fixed profiles: Superadministrador, Administrador, and Operador. Authorization to view screens or execute actions will depend exclusively on the role assigned to the account, preventing the custom assignment of individual permissions per user. The system must deny and block any attempt to access routes or functionalities for which the user's role does not possess explicit permissions. If a user attempts to enter a section or execute an action unauthorized for their role, the system will deny the request and display in the interface the message: "No tiene permisos para realizar esta acción".

FR-01-05: The system must prevent a single user account from maintaining more than one active session simultaneously across different devices or browsers. Upon the completion of a successful new login, the system will automatically invalidate the previous active session. If the user affected by this invalidation attempts to perform any action or navigation in the

previous session, the system will interrupt navigation, redirect them to the login screen, and display the message "Su sesión ha sido cerrada debido a un nuevo inicio de sesión en otro dispositivo".

FR-01-06: The system must monitor user interaction and automatically log out the user after a continuous 15-minute period of inactivity (absence of clicks or scrolls). Once this time has elapsed, the system will invalidate the session tokens, redirect the user to the login screen, and display the message "Su sesión ha caducado por inactividad".

FR-01-07: The system must execute an automatic scheduled differential backup every day at 18:00 hours (end of the workday) covering all critical system information, which includes the PostgreSQL database and server-stored files (generated quote PDF files). These backups must be stored encrypted in a secure external location or device. The system will manage an automated version rotation scheme that retains daily backups for the last 7 days, a weekly backup at the end of the last month, and a monthly backup during the last year, automatically purging files that exceed this retention schedule. In the event that the backup process fails or is interrupted, the system will immediately notify via email the users with the roles of Superadministrador and Administrador with the message "Fallo en la ejecución del respaldo automático del [Fecha]".

FR-01-08: The system must provide an exclusive module for the Administrador and Superadministrador roles that displays a detailed inventory of all available backup files. For each listed file, the interface must obligatorily show the exact creation date and time, the file size in megabytes, the backup type (daily, weekly, or monthly), and the integrity status resulting from a hash signature verification (SHA-256). The interface must allow applying search filters by any of these attributes (date range, size, backup type, and integrity status) and must allow authorized users to directly download any backup copy stored in the inventory.

FR-01-09: The system must provide users with the Administrador or Superadministrador role with a guided process for data restoration, which can be triggered directly from the backup inventory interface by selecting the desired backup copy. Before applying any changes to the production environment, the system must mandatorily execute an automated restoration test within an isolated environment. If the test in the isolated environment finishes successfully, the system will display in the interface the message: "Verificación exitosa. El respaldo es íntegro y apto para restauración" and enable the final confirmation option to proceed with restoration on the main system. If data corruption or errors are detected during the test, the system will abort the process, keep the main system intact, and notify on screen the message: "Error en la verificación del respaldo. Restauración cancelada".

FR-01-10: The system must provide a user management interface accessible exclusively to accounts with the Administrador or Superadministrador role. This interface must display a paginated list of all user accounts registered in the platform, obligatorily showing their basic identification data (full name, email address, assigned position, and system role), their current status (Activo or Inactivo), and the date of their last login. Additionally, the interface must offer search filters by name, email, position, role, and status, and incorporate visible options to trigger the creation, editing, and deactivation of each account.

FR-01-11: The system must allow accounts with the Administrador or Superadministrador role to register new user accounts from the user management interface. To complete registration, the system will obligatorily require the employee's basic identification data (full name, email address, and position), the selection of one of the three predefined roles in the platform (Superadministrador, Administrador, or Operador), and the assignment of an initial password, providing the option to enter it manually (which must comply with a minimum complexity criteria of 8 characters, at least one uppercase letter, one lowercase letter, one number, and one special character) or to request its automated random generation. The system will verify that the entered email is not previously registered; if it detects that the email is already in use, the system will prevent registration and notify on screen the message "El correo electrónico ya se encuentra registrado en el sistema". If unique, it will create the account in the Activo state and automatically send an email notification to the new user containing their credentials and initial access instructions.

FR-01-12: The system must allow accounts with the Administrador or Superadministrador role to modify existing user account information from the user management interface. The system will allow updating the full name, position, assigned role, and access password, offering for the latter the option to enter it manually or generate a new automated random value. For traceability and security reasons, the email address associated with the account cannot be modified. Upon saving changes, if the password was updated, the system will notify the user via email with their new credentials; likewise, role updates will immediately apply the new permissions and restrictions corresponding to the account.

FR-01-13: The system must allow accounts with the Administrador or Superadministrador role to modify the status of a user account between "Activo" and "Inactivo" from the user management interface, preventing the physical deletion of records in the database to preserve historical traceability. Upon deactivating an account, the system will immediately invalidate any active session the user maintains on the platform and prevent subsequent logins, showing the message "Su cuenta se encuentra desactivada. Contacte al administrador". Likewise, the system must allow reactivating a previously deactivated account, restoring its access according to its assigned role.

Module 2: Commercial Management and Virtual Assistant

FR-02-01: The system must provide a centralized interface for searching and managing the product catalog, accessible to accounts with the Administrador, Superadministrador, or Operador role (the latter with read-only access). The interface must display a paginated list of all products registered in the platform, showing mandatorily for each item its unique code, description or name, unit of measure, and its current status (Activo or Inactivo). Likewise, the interface must incorporate search filters by code, description, unit of measure, and status, and include visible options to trigger the registration of new products, the editing of existing ones, and their status updates.

FR-02-02: The system must allow accounts with the Administrador or Superadministrador role to register new products within the central catalog from the catalog interface. Upon registration, the system will mandatorily require the following attributes: unique product code, description or name, and unit of measure. The system will verify that the unique product code does not previously exist in the database; if a duplicate is detected, it will prevent

registration and display in the interface the message "El código de producto ya se encuentra registrado en el catálogo". Once the record is saved successfully, the product will immediately become available in the Activo state for query and selection within the platform.

FR-02-03: The system must allow accounts with the Administrador or Superadministrador role to modify the information of existing products from the central product catalog interface. The system will allow updating the product description or name, and unit of measure. For reasons of referential integrity and traceability in system records, the unique product code cannot be modified once created.

FR-02-04: The system must allow accounts with the Administrador or Superadministrador role to modify the status of a product between "Activo" and "Inactivo" from the central product catalog interface, preventing the physical deletion of the record in the database to preserve the integrity of historical data. Upon marking a product as "Inactivo", the system will immediately prevent its selection in the creation of new quotes on the platform, keeping it visible only in the histories of operations created before its deactivation. Likewise, the system must allow reactivating a previously inactivated product to enable its use again in creating new transactions.

FR-02-05: The system must integrate a conversational assistance interface (chatbot) on the company's public web portal99, powered by the DeepSeek-V3 model, allowing potential customers to request and receive automated quotes containing an ideal price range estimated by the AI based on historical prices used in recent quotes. To process the request, the chatbot will mandatorily require the applicant's contact details: full name, email address, and phone number. Once the required products and quantities are collected, the system will automatically calculate the corresponding estimated price ranges, generate a PDF document with the detailed quote (including item breakdown, unit prices, totals, and validity period), and offer the customer the option to download it directly or receive it via email immediately. Likewise, the system will automatically log the generated quote within an internal tracking module, leaving it available in the Pendiente state for subsequent management.

FR-02-06: The system must calculate and display an automated suggestion of the estimated dispatch date when registering or elaborating quotes and orders in the platform. The calculation of this estimate will be generated by applying the historical median of actual delivery times recorded in the system for the set of products and volumes included in the request. In case there are insufficient historical records to determine the estimate for a specific product or volume combination, the system will display in the interface the message: "Sin datos históricos suficientes para estimar la fecha de despacho" and require manual entry of the date by the user. The suggested or entered date will be visible in the interface to support the operational management of users with the Operador, Administrador, or Superadministrador role, allowing them to confirm the suggestion or manually adjust the final dispatch date according to negotiation criteria or agreements.

FR-02-07: The system must provide a customer management interface accessible to accounts with the Operador, Administrador, or Superadministrador role that displays a paginated list of all customers registered on the platform. The screen must show for each record the business name or full name, Tax Identification Number (NIT) or identity document number, main commercial contact, and status (Activo or Inactivo). Likewise, the interface

must offer search filters by NIT or document number, business name or full name, commercial contact, and status, and incorporate direct access from each row to navigate to the unified profile sheet of the selected customer, as well as visible options to trigger the registration of new customers, modification of their data, and annulment or deactivation of existing records.

FR-02-08: The system must provide a detailed query screen for each customer registered on the platform, accessible from the customer management interface to accounts with the Operador, Administrador, or Superadministrador role and indexed directly using their Tax Identification Number (NIT) or identity document number. The interface must consolidate and display in a unified view general and contact customer information (business name or full name, NIT or document number, address, contact phone numbers, and emails), the paginated chronological history of their quotes and orders, and the log of internal commercial tracking notes.

FR-02-09: The system must allow accounts with the Operador, Administrador, or Superadministrador role to register new customers on the platform from the customer management interface. To complete registration, the system will require selecting the customer type (natural or legal person) and mandatorily capturing the business name or full name, Tax Identification Number (NIT) or identity document number, tax or home address, primary phone number, and email address. The system will automatically validate that the entered NIT or document number does not previously exist in the database; in case of duplication, it will prevent creation and display an alert message on screen indicating that the document is already registered. Once saved successfully, the new customer will remain active by default and available for generating quotes, orders, and queries on the platform.

FR-02-10: The system must allow accounts with the Operador, Administrador, or Superadministrador role to modify existing customer information from the customer management interface. The system will allow updating the business name or full name, address, primary phone number, and contact email address. For reasons of referential integrity and traceability in system records, neither person type (natural or legal) nor Tax Identification Number (NIT) or identity document number may be modified once the record is created. Upon saving changes, updated information will be reflected immediately in the customer's unified profile sheet and in future operations generated on the platform, keeping intact the data recorded in previous transactions.

FR-02-11: The system must allow accounts with the Operador, Administrador, or Superadministrador role to deactivate or annul registered customers from the customer management interface. The operation will consist of a logical flag changing the customer status from Activo to Inactivo, without physically deleting the record or its associated data from the database to preserve historical traceability and integrity of previously generated quotes and orders. Upon entering the Inactivo state, the system will prevent the customer from being selected or associated with the elaboration of new quotes or orders on the platform; however, their unified profile and past transactions will remain queryable in read-only mode for audit and administrative management purposes.

FR-02-12: The system must provide a quote management interface accessible to accounts with the Administrador or Superadministrador role that displays a paginated list of all

preliminary quotes generated automatically by the virtual assistant. The screen must show for each record the date and time of issuance, applicant contact details, quoted products, estimated total amount, and current status (Pendiente de revisión, Aprobada, Modificada, or Rechazada). Likewise, the interface must offer search filters corresponding to each of these fields (date and time of issuance, applicant contact details, quoted products, estimated total amount, and status), and incorporate in each row visible options to execute actions to approve, modify, or reject the corresponding preliminary quote.

FR-02-13: The system must allow accounts with the Administrador or Superadministrador role to approve a preliminary quote generated by the virtual assistant directly from the quote management interface. Upon executing the approval action on a record in the Pendiente de revisión or Modificada status, the system will change its status to Aprobada, log the date, time, and user responsible for the status change in the system log, and automatically send a notification email to the applicant attaching the formal official quote document, after which the record will remain immutable.

FR-02-14: The system must allow accounts with the Administrador or Superadministrador role to modify a preliminary quote generated by the virtual assistant from the quote management interface when it is in the Pendiente de revisión or Modificada status. The modification interface will allow adjusting quantities, adding or removing products, modifying applicable unit prices, commercial conditions, or estimated delivery time. Upon saving changes, the system will automatically recalculate the estimated total amount, update or maintain the quote status as Modificada, and log the date, time, user responsible, and detail of modifications in the system log.

FR-02-15: The system must allow accounts with the Administrador or Superadministrador role to reject a preliminary quote generated by the virtual assistant from the quote management interface when it is in the Pendiente de revisión or Modificada status. Upon executing the rejection action, the system will mandatorily request entry of a reason or justification for rejection, change the proposal status to Rechazada, and log the date, time, user responsible, and entered reason in the system log. Likewise, the system will automatically send a notification email to the applicant informing them of the request rejection along with the corresponding justification, leaving the quote immutable for future modifications.

FR-02-16: The system must automatically incorporate commercial orders into an administrative processing queue immediately after they reach the Aprobada status. This automated process must log the date, time, and unique order identifier in the queue, ensuring that the record remains available in waiting mode for subsequent consolidation and mass export to the accounting system, preventing orders in preliminary, pending review, or rejected states from entering this flow.

FR-02-17: The system must provide an interface accessible to accounts with the Operador, Administrador, or Superadministrador role displaying the list of approved commercial orders available in the administrative processing queue. The screen must show for each record the date and time of approval, customer, products, total amount, and export status, incorporating search filters by date range, customer, and order identifier, as well as multi-selection controls (checkboxes) allowing the user to select records to include in an export batch.

FR-02-18: The system must allow accounts with the Operador, Administrador, or Superadministrador role to process the multi-selection of orders from the export queue to generate and download a single unidirectional flat file. This file must be structured under the technical specifications and strict format required for batch loading into the Siigo Pyme accounting system, logging the date, time, user responsible, and batch detail in the system log, and marking exported orders to exclude them from the active queue and prevent duplication in subsequent downloads.

FR-02-19: The system must provide a work order management interface accessible to accounts with the Operador, Administrador, or Superadministrador role that displays a paginated list of all Work Orders (O.T.) registered on the platform. The screen must show for each record the O.T. number, customer information, details of associated items, agreed delivery date, and current status (Pendiente, Registrada en Siigo, Conciliada, or Anulada). Likewise, the interface must offer search filters corresponding to each of these fields (O.T. number, customer, item details, agreed date, and status), visible access to register new work orders, and incorporate in each row corresponding options to edit or annul the selected work order.

FR-02-20: The system must allow accounts with the Operador, Administrador, or Superadministrador role to manually register a new Work Order (O.T.) on the platform. The interface will mandatorily require customer selection or assignment, details of associated items or products with their respective quantities, and the agreed delivery date. Upon saving information, the system will automatically assign a unique, consecutive O.T. number, set its initial status as Pendiente, and log the date, time, and creator user in the system log.

FR-02-21: The system must allow accounts with the Operador, Administrador, or Superadministrador role to modify information of an existing Work Order (O.T.) from the management interface when it is in the Pendiente or Registrada en Siigo status. The editing interface will allow adjusting customer selection, adding, modifying, or removing items with their quantities, updating agreed delivery date, and modifying operational order status. Upon saving changes, the system will update record information, keep original assigned O.T. number immutable, and log date, time, responsible user, and modified data detail in system log.

FR-02-22: The system must allow accounts with the Operador, Administrador, or Superadministrador role to annul a Work Order (O.T.) registered on the platform directly from management interface when it is in Pendiente or Registrada en Siigo status. Upon executing annulment action, system will mandatorily require entry of justification or reason for annulment, update order status to Anulada, and log date, time, responsible user, and entered reason in system log.

FR-02-23: The system must control lifecycle evolution of each Work Order (O.T.), allowing accounts with Operador, Administrador, or Superadministrador role to update its operational status solely following a strict sequential flow starting in Pendiente, advancing to Registrada en Siigo, and culminating in Conciliada. System will prevent any jump between non-consecutive statuses in said flow, validate that transition to Anulada status is allowed only from Pendiente or Registrada en Siigo statuses, and log each status change alongside date, time, and responsible user in system log to guarantee process traceability.

Non-Functional Requirements

Category 01: Security and Auditing

NFR-01-01: The system must guarantee that all communication between the web client and the server travels encrypted via the HTTPS protocol, using at least TLS 1.2 (with TLS 1.3 as the preferred version) and a valid, current digital certificate. Any request made over HTTP must be automatically redirected to HTTPS, and the server must include the HTTP Strict Transport Security (HSTS) header to prevent the browser from establishing unencrypted connections again. This requirement applies to all views, API endpoints, and system resources, including the virtual assistant on the public portal, and no credentials, tokens, or sensitive data may be transmitted as part of the URL.

NFR-01-02: The system must store user passwords in the PostgreSQL database exclusively as hashes, using the bcrypt algorithm with a minimum cost factor of 12 and a unique salt generated independently for each password. Under no circumstances will passwords be stored in plain text or with reversible encryption, and the hash cannot be returned in API responses nor recorded in audit logs, log files, or error messages. Login validation will always be performed by comparing the calculated hash with the stored hash.

NFR-01-03: The system must maintain an unalterable, detailed, and chronological audit log of all critical actions executed on the platform, such as reconciliations, role and account status modifications, user creation or deactivation, backup operations, and all login attempts, both successful and failed. Each entry must retain, at a minimum, the exact date and time with time zone, the responsible user and their role, the source IP address, the type of action, the affected entity, the previous and new values where applicable, and the result of the operation. The log must operate under an insert-only scheme, with update and delete permissions revoked at the database level, ensuring that no user profile, including Superadministrador, can modify or delete its entries, and it must be retained for a minimum period of 5 years.

NFR-01-04: The system must encrypt all backups of the database and server files under the AES-256 standard before they leave the origin server, ensuring that no backup exists or is transferred in plain text. Encryption keys must be stored separately from the copies they protect and from the external location where these are saved, never being included in the source code or repositories, and access to them must be limited to authorized personnel. The transfer to the external location or device must be carried out through a secure channel, and the system must be able to decrypt and restore the information only by using the corresponding key.

Category 02: Performance and Efficiency

NFR-02-01: The system must ensure a fluid user experience, guaranteeing that standard navigation requests and information queries are processed and fully rendered in a maximum time of 2 seconds. This response time must be met for at least 95% of requests, measured from the moment the user executes the action until the information is visible on the screen, under the company's local network and with a load of up to 20 concurrent users. To achieve this, listings must be presented in a paginated manner and database queries must be

supported by appropriate indexes, without loading unnecessary volumes of information in a single response.

Category 03: Usability

NFR-03-01: The system's graphical interface must be responsive and adapt automatically to the different screen resolutions of the company's computers and laptops, from 1366 x 768 pixels onwards, without losing functionality, hiding essential actions or information, or requiring horizontal scrolling to view content. Large tables and lists must reorganize or adjust their presentation to the available width, and the design must react correctly when resizing the browser window without needing to reload the page.

NFR-03-02: The system's graphical interface must be intuitive, so that a user without prior training or assistance can complete any operational workflow in less than five minutes. This condition must be verified through usability testing with at least five representative users from the different roles. To achieve this, the interface must maintain consistent navigation and element layout across all modules, use clear labels and messages in Spanish, indicate precisely how to correct validation errors, and request explicit confirmation before executing high-impact actions.

Category 04: Availability and Reliability

NFR-04-01: The system must operate stably on the company's local network, guaranteeing a minimum availability of 99% during the daily workday. Over an 8-hour workday across 22 business days per month, this equals a cumulative downtime of no more than 1 hour and 45 minutes per month. The services composing the system must restart automatically in the event of an unexpected crash, and maintenance, update, or deployment tasks must be scheduled outside of business hours. In the event of a failure, the system must become fully operational again within a maximum time of 30 minutes, without losing previously confirmed transactions.

Category 05: Architecture and Maintainability

NFR-05-01: The backend source code must mandatorily be structured through the implementation of the Strategy, Factory, Observer, and Repository design patterns, and each must be accompanied by formal documentation. Strategy will be used to encapsulate interchangeable algorithms, Factory to centralize object creation, Observer to propagate events between decoupled components, and Repository to isolate database access from business logic. The documentation for each pattern must include its purpose, the justification for its use, a class diagram, and the location of its implementation in the code, and its compliance must be verified during code reviews.

NFR-05-02: All new code or functionality must pass through an automated validation pipeline configured in GitHub Actions, which will execute on every integration of changes and must successfully complete project compilation along with a suite of unit and integration tests, executed exclusively on synthetic data and never on real production information. The pipeline must achieve a minimum test coverage of 70% on the backend code. Any failure in any of its stages must block deployment to production, and the main branch must be protected so that it only receives changes whose pipeline has completed successfully.

Category 06: Interoperability and Compatibility

NFR-06-01: The system interface must be accessible and fully functional from any computer via a standard web browser, on current stable versions of Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. Its operation must be smooth without requiring the installation of additional programs, plugins, or extensions, and independent of the operating system (Windows, macOS, or Linux) and special hardware capabilities, requiring only a conventional office computer with a connection to the local network.

User Stories

Module 1: Information Security

US 1.1: As a system user, **I want** to log in with my email address and password **so that** I can access the platform and carry out the operational activities corresponding to my position.

- **Criterion 1.1.1: Given** that the user is registered, active, and has a linked authenticator app, **when** they enter their correct email address and password, **then** the system generates a temporary pre-authentication token and directs them to the 6-digit code verification screen.
- **Criterion 1.1.2: Given** that the user entered correct credentials, **when** the system validates the password, **then** it does not issue final JWT tokens nor grant access to any module until 2FA verification is completed.
- **Criterion 1.1.3: Given** that the user enters an unregistered email, **when** they attempt to log in, **then** the system denies access and displays the message "Correo electrónico o contraseña incorrectos".
- **Criterion 1.1.4: Given** that the user enters a registered email with an incorrect password, **when** they attempt to log in, **then** the system denies access and displays the same message "Correo electrónico o contraseña incorrectos", without indicating which of the two inputs failed.
- **Criterion 1.1.5: Given** that the user leaves the email or password fields empty, **when** they attempt to log in, **then** the system blocks submission and indicates that both fields are mandatory.
- **Criterion 1.1.6: Given** that the user is registered but does not yet have a linked authenticator app, **when** they enter correct credentials, **then** the system directs them to the initial pairing process.
- **Criterion 1.1.7: Given** that the account is in the "Inactivo" state, **when** the user enters correct credentials, **then** the system denies access and displays the message "Su cuenta se encuentra desactivada. Contacte al administrador".

US 1.2: As a system user without a configured authenticator, **I want** to link an external authenticator application using a QR code **so that** two-step verification is activated for my future logins.

- **Criterion 1.2.1: Given** that the user validated their credentials and does not have a linked authenticator, **when** they access pairing, **then** the system displays a unique QR code along with its alphanumeric secret key in text, valid for a maximum of 5 minutes.

- **Criterion 1.2.2: Given** that the QR and secret key are valid, **when** the user registers them in their authenticator app and enters the generated TOTP code, **then** the system validates the code, confirms the link, and activates mandatory two-factor authentication for future logins.
- **Criterion 1.2.3: Given** that the user attempts to confirm pairing, **when** the entered code is incorrect, **then** the system rejects the attempt, keeps pairing pending, and allows retries while the QR remains valid.
- **Criterion 1.2.4: Given** that 5 minutes elapsed without completing pairing, **when** the user attempts to confirm it, **then** the system invalidates the QR and secret key and displays the message "El tiempo de vinculación ha expirado. Por favor, genere un nuevo código".
- **Criterion 1.2.5: Given** that the QR code expired, **when** the user explicitly requests generating a new code, **then** the system issues a new QR and secret key, restarting the 5-minute validity window.
- **Criterion 1.2.6: Given** that the user already has a linked authenticator, **when** they attempt to access initial pairing or generate a new QR, **then** the system blocks the action and indicates that an active pairing already exists.

US 1.3: As a system user, **I want** to enter the 6-digit temporary code from my authenticator app after my password **so that** I can confirm my identity and access the platform.

- **Criterion 1.3.1: Given** that the user passed credential validation and holds a valid pre-authentication token, **when** they enter a correct and active 6-digit TOTP code, **then** the system issues access and refresh JWT tokens and allows them to enter the platform.
- **Criterion 1.3.2: Given** that the user is in 2FA verification, **when** they enter an incorrect or expired code, **then** the system denies access and displays the message "Código de verificación incorrecto o expirado".
- **Criterion 1.3.3: Given** that the user is in 2FA verification, **when** they enter a code with an invalid format (less or more than 6 digits, or non-numeric characters), **then** the system rejects the attempt without consuming verification and indicates that the code must contain 6 digits.
- **Criterion 1.3.4: Given** that the user has not completed 2FA verification, **when** they attempt to access a route or module directly, **then** the system denies access and redirects them to login.

US 1.4: As an Administrador, **I want** access to modules, views, and actions to depend exclusively on each account's role (Superadministrador, Administrador, or Operador) **so that** each person only executes operations corresponding to their role.

- **Criterion 1.4.1: Given** that the user confirmed authentication, **when** they enter the platform, **then** the system enables in the interface only those modules and functionalities permitted for their role.
- **Criterion 1.4.2: Given** that a user attempts to access an unauthorized section for their role via navigation or typing the route directly, **when** the system evaluates the request, **then** it denies it and displays the message "No tiene permisos para realizar esta acción".

- **Criterion 1.4.3: Given** that a user attempts to execute an unauthorized operation for their role, **when** the system evaluates the request, **then** it blocks it even if the option is not visible in the interface and displays the message "No tiene permisos para realizar esta acción".
- **Criterion 1.4.4: Given** that the Administrador manages account roles, **when** they consult available role options, **then** the system offers only the three fixed profiles (Superadministrador, Administrador, and Operador) and does not permit custom permissions per user.
- **Criterion 1.4.5: Given** that an account role was modified, **when** the system applies the change, **then** new permissions and restrictions apply immediately for that account.

US 1.5: As a system user, **I want** my account to maintain only one active session at a time **so that** unauthorized simultaneous access using my credentials is prevented.

- **Criterion 1.5.1: Given** that the user maintains an active session on a device or browser, **when** a new successful login is completed with the same account on another device or browser, **then** the system automatically invalidates the previous session.
- **Criterion 1.5.2: Given** that the previous session was invalidated by a new login, **when** the user attempts to perform an action or navigate within it, **then** the system interrupts navigation, redirects them to the login screen, and displays the message "Su sesión ha sido cerrada debido a un nuevo inicio de sesión en otro dispositivo".
- **Criterion 1.5.3: Given** that a new login occurs, **when** the system invalidates the previous session, **then** the JWT tokens of that session cease to be valid for any subsequent request.

US 1.6: As a system user, **I want** my session to automatically close after a period of inactivity **so that** another person cannot use my account if I leave my equipment unattended.

- **Criterion 1.6.1: Given** that the user has an active session, **when** 15 continuous minutes elapse without interaction (no clicks or scrolling), **then** the system automatically closes the session and invalidates its tokens.
- **Criterion 1.6.2: Given** that the session closed due to inactivity, **when** the system terminates it, **then** it redirects the user to the login screen and displays the message "Su sesión ha caducado por inactividad".
- **Criterion 1.6.3: Given** that the user has an active session, **when** they click or scroll before 15 minutes pass, **then** the system resets the inactivity timer.
- **Criterion 1.6.4: Given** that the session was closed due to inactivity, **when** the user attempts to perform an action using previous tokens, **then** the system rejects the request and requires a new login.

US 1.7: As an Administrador, **I want** the system to automatically generate an encrypted differential backup at the end of each workday **so that** critical company information is protected against loss or failures.

- **Criterion 1.7.1: Given** that the system is operating, **when** it reaches 18:00 hours each day, **then** it automatically initiates a differential backup.

- **Criterion 1.7.2: Given** that the backup executes, **when** the process completes, **then** the file includes the PostgreSQL database, and generated quote PDF files.
- **Criterion 1.7.3: Given** that the backup was generated, **when** the system stores it, **then** it saves it encrypted in a secure external location or device.
- **Criterion 1.7.4: Given** that the backup was stored, **when** the system registers it, **then** it calculates its hash signature (SHA-256) and associates it with the file for subsequent integrity verification.

US 1.8: As an Administrador, **I want** the system to automatically rotate and purge backup versions according to a retention policy **so that** necessary history is preserved without saturating storage.

- **Criterion 1.8.1: Given** that the system generates daily backups, **when** a new backup completes, **then** it retains daily backups for the last 7 days.
- **Criterion 1.8.2: Given** that the system completes a weekly cycle, **when** the week ends, **then** it retains one weekly backup for each week of the last month.
- **Criterion 1.8.3: Given** that the system completes a monthly cycle, **when** the month ends, **then** it retains one monthly backup for each month of the last year.
- **Criterion 1.8.4: Given** that backup files exist exceeding the retention schedule, **when** the system executes rotation, **then** it purges them automatically.

US 1.9: As an Administrador, **I want** to receive an email notification when automated backup fails **so that** I can act in time and avoid being left without a backup copy.

- **Criterion 1.9.1: Given** that scheduled backup fails or is interrupted, **when** the system detects the error, **then** it immediately sends an email to all active accounts with the Administrador and Superadministrador roles.
- **Criterion 1.9.2: Given** that failure notification is sent, **when** the user receives it, **then** the message indicates "Fallo en la ejecución del respaldo automático del [Fecha]" with the date corresponding to the failed backup.
- **Criterion 1.9.3: Given** that backup failed, **when** the system logs the event, **then** the failed backup does not appear in inventory as a copy available for download or restoration.

US 1.10: As an Administrador, **I want** to consult the inventory of available backup copies **so that** I can verify that the system backs up information and complies with the retention policy.

- **Criterion 1.10.1: Given** that the Administrador accesses the backup module, **when** inventory loads, **then** the system lists all available backup files.
- **Criterion 1.10.2: Given** that inventory is visible, **when** the Administrador reviews each record, **then** the system displays the exact creation date and time, size in megabytes, backup type (daily, weekly, or monthly), and integrity status.
- **Criterion 1.10.3: Given** that the system verifies backup integrity, **when** it compares its hash signature (SHA-256) against the one recorded upon creation, **then** integrity status reflects the comparison result.
- **Criterion 1.10.4: Given** that no backups have been generated yet, **when** the Administrador accesses inventory, **then** the system displays a message indicating no copies are available.

- **Criterion 1.10.5: Given** that a user with the Operador role attempts to access the backup module, **when** they execute the action, **then** the system denies access with the message "No tiene permisos para realizar esta acción".

US 1.11: As an Administrador, **I want** to filter the backup inventory by various attributes **so that** I can quickly locate the copy I need.

- **Criterion 1.11.1: Given** that inventory is visible, **when** the Administrador filters by date range, **then** the system displays only backups created within that range.
- **Criterion 1.11.2: Given** that inventory is visible, **when** the Administrador filters by size, **then** the system displays only backups meeting that size criterion.
- **Criterion 1.11.3: Given** that inventory is visible, **when** the Administrador filters by backup type (daily, weekly, or monthly), **then** the system displays only backups of the selected type.
- **Criterion 1.11.4: Given** that inventory is visible, **when** the Administrador filters by integrity status, **then** the system displays only backups with that status.
- **Criterion 1.11.5: Given** that the Administrador applies multiple filters simultaneously, **when** they execute the search, **then** the system displays only backups meeting all combined criteria.
- **Criterion 1.11.6: Given** that no backup meets applied filters, **when** the system executes search, **then** it displays a message indicating no results were found.

US 1.12: As an Administrador, **I want** to directly download a stored backup copy **so that** I can keep it outside the system when needed.

- **Criterion 1.12.1: Given** that the Administrador views inventory, **when** they select backup download, **then** the system starts downloading the corresponding file.
- **Criterion 1.12.2: Given** that a backup is downloading, **when** download completes, **then** the file is delivered in its encrypted form, exactly as stored.
- **Criterion 1.12.3: Given** that a user with the Operador role attempts to download a backup, **when** they execute the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 1.13: As an Administrador, **I want** the system to automatically test a backup in an isolated environment before restoring it **so that** I can ensure it is not corrupt and avoid risking operations.

- **Criterion 1.13.1: Given** that the Administrador views backup inventory, **when** they select a copy and choose restore, **then** the system begins a guided process and mandatorily executes an automated restoration test in an isolated environment.
- **Criterion 1.13.2: Given** that the test runs in an isolated environment, **when** the process takes place, **then** the main production system remains entirely unchanged.
- **Criterion 1.13.3: Given** that the isolated environment test completes successfully, **when** the system displays the result, **then** it presents the message "Verificación exitosa. El respaldo es íntegro y apto para restauración".
- **Criterion 1.13.4: Given** that the test detects data corruption or errors, **when** the system evaluates the result, **then** it aborts the process, keeps the main system intact, and displays the message "Error en la verificación del respaldo. Restauración cancelada".

- **Criterion 1.13.5: Given** that verification failed, **when** the Administrador reviews inventory, **then** the final restoration confirmation option is not enabled for that backup.
- **Criterion 1.13.6: Given** that a user with the Operador role attempts to initiate restoration, **when** they execute the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 1.14: As an Administrador, **I want** to explicitly confirm restoration of a verified backup over the main system **so that** I can recover information in a controlled manner.

- **Criterion 1.14.1: Given** that verification in the isolated environment was successful, **when** the system displays the result, **then** it enables the final confirmation option to proceed with restoration.
- **Criterion 1.14.2: Given** that final confirmation is enabled, **when** the Administrador triggers it, **then** the system applies backup data over the production environment.
- **Criterion 1.14.3: Given** that verification was successful, **when** the Administrador chooses not to confirm or cancels the process, **then** the system keeps the production environment intact.

US 1.15: As an Administrador, **I want** to consult a paginated list of user accounts **so that** I know who has access to the platform, with what role, and in what status.

- **Criterion 1.15.1: Given** that the Administrador accesses user management, **when** the list loads, **then** the system displays registered accounts in a paginated format.
- **Criterion 1.15.2: Given** that the list is visible, **when** the Administrador reviews each record, **then** the system displays full name, email address, position, role, status (Activo or Inactivo), and last access date.
- **Criterion 1.15.3: Given** that a user has never logged in, **when** their record is displayed, **then** the last access field indicates no logins recorded.
- **Criterion 1.15.4: Given** that the list is visible, **when** the Administrador reviews account actions, **then** the system displays options to create, edit, and deactivate.
- **Criterion 1.15.5: Given** that a user with the Operador role attempts to access user management, **when** they execute the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 1.16: As an Administrador, **I want** to filter and search user accounts **so that** I can quickly find the account I need to manage.

- **Criterion 1.16.1: Given** that the user list is visible, **when** the Administrador searches by name, **then** the system displays accounts whose name matches the search.
- **Criterion 1.16.2: Given** that the user list is visible, **when** the Administrador searches by email, **then** the system displays matching accounts.
- **Criterion 1.16.3: Given** that the user list is visible, **when** the Administrador filters by position, role, or status, **then** the system displays only accounts meeting the selected criterion.
- **Criterion 1.16.4: Given** that the Administrador combines multiple filters, **when** they execute search, **then** the system displays only accounts meeting all criteria.
- **Criterion 1.16.5: Given** that no account meets filters, **when** search executes, **then** the system displays a message indicating no results were found.

US 1.17: As an Administrador, **I want** to register new user accounts **so that** I can grant platform access to employees who require it.

- **Criterion 1.17.1: Given** that the Administrador is in user management, **when** they select to create a new account, **then** the system displays a form with mandatory fields: full name, email address, position, role, and initial password.
- **Criterion 1.17.2: Given** that the Administrador fills out the form, **when** selecting the role, **then** the system offers only Superadministrador, Administrador, and Operador.
- **Criterion 1.17.3: Given** that the Administrador omits a mandatory field, **when** attempting to save, **then** the system blocks the action and indicates missing fields.
- **Criterion 1.17.4: Given** that the Administrador enters an already registered email, **when** attempting to save, **then** the system prevents registration and displays the message "El correo electrónico ya se encuentra registrado en el sistema".
- **Criterion 1.17.5: Given** that the Administrador enters an email with an invalid format, **when** attempting to save, **then** the system blocks the action and indicates invalid email format.
- **Criterion 1.17.6: Given** that the Administrador decides to enter password manually, **when** typing it, **then** the system requires a minimum of 8 characters with at least one uppercase letter, one lowercase letter, one number, and one special character, rejecting non-compliant passwords.
- **Criterion 1.17.7: Given** that the Administrador requests automated password generation, **when** the system generates it, **then** the created value meets defined complexity criteria.
- **Criterion 1.17.8: Given** that data is valid and email is unique, **when** the Administrador saves, **then** the system creates the account in "Activo" state.
- **Criterion 1.17.9: Given** that the account was created, **when** registration completes, **then** the system automatically sends an email to the new user containing their credentials and initial access instructions.

US 1.18: As an Administrador, **I want** to modify user account details **so that** I can keep information and permissions updated.

- **Criterion 1.18.1: Given** that the Administrador is on the user list, **when** selecting to edit an account, **then** the system displays the form populated with current account data.
- **Criterion 1.18.2: Given** that edit form is open, **when** the Administrador reviews it, **then** it permits editing full name, position, and role, displaying email as a non-editable field.
- **Criterion 1.18.3: Given** that the Administrador leaves a mandatory field empty, **when** attempting to save, **then** the system blocks the action and indicates missing fields.
- **Criterion 1.18.4: Given** that the Administrador modifies data with valid info, **when** saving changes, **then** the system updates account information and confirms operation.
- **Criterion 1.18.5: Given** that the Administrador changes account role, **when** saved, **then** new permissions and restrictions apply immediately for that account.
- **Criterion 1.18.6: Given** an attempt to alter account email, **when the system** processes a request, **then** it rejects it and preserves original email.

US 1.19: As an Administrador, **I want** to update a user account password **so that** I can reset access when necessary.

- **Criterion 1.19.1: Given** that the edit form is open, **when** the Administrador chooses to change password, **then the system** offers entering it manually or generating a new random one.
- **Criterion 1.19.2: Given** that the Administrador enters password manually, **when** typing it, **then the system** requires a minimum of 8 characters with at least one uppercase, one lowercase, one number, and one special character, rejecting non-compliant values.
- **Criterion 1.19.3: Given** that the Administrador requests a random password, **when the system** generates it, **then the value** meets defined complexity criteria.
- **Criterion 1.19.4: Given** that password was updated, **when** the Administrador saves changes, **then the system** notifies the user via email with new credentials.

US 1.20: As an Administrador, **I want** to deactivate user accounts without deleting them **so that** I can prevent access for users who no longer require it while preserving historical traceability.

- **Criterion 1.20.1: Given** that the Administrador is on the user list, **when** selecting an active account, **then the system** enables the deactivation option.
- **Criterion 1.20.2: Given** that the Administrador confirms deactivation, **when the system** executes it, **then** it changes account status to "Inactivo" and preserves database records.
- **Criterion 1.20.3: Given** that the account has an active session, **when the system** processes deactivation, **then** it immediately invalidates that session.
- **Criterion 1.20.4: Given** that an account is in "Inactivo" state, **when** user attempts login, **then** system denies access and displays message "Su cuenta se encuentra desactivada. Contacte al administrador".

US 1.21: As an Administrador, **I want** to reactivate previously deactivated user accounts **so that** I can restore access to users who require it again.

- **Criterion 1.21.1: Given** that the Administrador is on the user list, **when** selecting an inactive account, **then the system** enables the reactivation option.
- **Criterion 1.21.2: Given** that the Administrador confirms reactivation, **when the system** executes it, **then** it changes account status to "Activo" and restores access according to the assigned role.
- **Criterion 1.21.3: Given** that account was reactivated, **when the user** logs in with credentials, **then** normal two-factor authentication flow resumes.

Module 2: Commercial Management and Virtual Assistant

US 2.1: As an Operador, **I want** to consult the product catalog **so that** I can know the company's current offering and available products.

- **Criterion 2.1.1: Given** that the user accesses the product catalog, **when** the interface loads, **then** the system displays a paginated list with all registered products.

- **Criterion 2.1.2: Given** that the list is visible, **when** the user reviews each record, **then** the system displays the unique code, description or name, unit of measure, and status (Activo or Inactivo).
- **Criterion 2.1.3: Given** that a user with the Administrador or Superadministrador role consults the catalog, **when** reviewing interface options, **then** the system displays actions to register product, edit product, and update status.
- **Criterion 2.1.4: Given** that a user with the Operador role consults the catalog, **when** reviewing the interface, **then** the system offers read-only access without actions to register, edit, or change status.
- **Criterion 2.1.5: Given** that no registered products exist, **when** the user accesses the catalog, **then** the system displays a message indicating that no products are available.

US 2.2: As an Operador, **I want** to filter products in the catalog **so that** I can quickly locate the product I need.

- **Criterion 2.2.1: Given** that the catalog list is visible, **when** the user filters by code, **then** the system displays only products whose code matches the search.
- **Criterion 2.2.2: Given** that the catalog list is visible, **when** the user filters by description, **then** the system displays only products whose description matches the search.
- **Criterion 2.2.3: Given** that the catalog list is visible, **when** the user filters by unit of measure or status, **then** the system displays only products meeting the selected criterion.
- **Criterion 2.2.4: Given** that the user applies multiple filters simultaneously, **when** executing search, **then** the system displays only products meeting all criteria.
- **Criterion 2.2.5: Given** that no product meets applied filters, **when** search executes, **then** the system displays a message indicating no results were found.

US 2.3: As an Administrador, **I want** to register new products in the central catalog **so that** I can digitize and centralize the company's offering.

- **Criterion 2.3.1: Given** that the Administrador is in the catalog, **when** selecting to register a product, **then** the system displays a form with mandatory fields: unique code, description or name, and unit of measure.
- **Criterion 2.3.2: Given** that the Administrador omits a mandatory field, **when** attempting to save, **then** the system blocks the action and indicates missing fields.
- **Criterion 2.3.3: Given** that the Administrador enters an existing code, **when** attempting to save, **then** the system prevents registration and displays the message "El código de producto ya se encuentra registrado en el catálogo".
- **Criterion 2.3.4: Given** that data is valid and the code is unique, **when** the Administrador saves, **then** the system creates the product in "Activo" state and makes it available immediately for consultation and selection.
- **Criterion 2.3.5: Given** that a user with the Operador role attempts to register a product, **when** executing the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 2.4: As an Administrador, **I want** to modify the description and unit of measure of a product **so that** I can keep its information updated in the catalog.

- **Criterion 2.4.1: Given** that the Administrador is in the catalog, **when** selecting edit a product, **then** the system displays the form populated with current product data.
- **Criterion 2.4.2: Given** that the edit form is open, **when** the Administrador reviews it, **then** it permits modifying description or name and unit of measure, displaying the unique code as a non-editable field.
- **Criterion 2.4.3: Given** an attempt to alter an existing product's code, **when** the system processes the request, **then** it rejects it and preserves the original code.
- **Criterion 2.4.4: Given** that the Administrador leaves a mandatory field empty, **when** attempting to save, **then** the system blocks the action and indicates missing fields.
- **Criterion 2.4.5: Given** that the Administrador modifies data with valid info, **when** saving changes, **then** the system updates the product, reflects it in the catalog, and keeps intact data for products already recorded in previous quotes and orders.
- **Criterion 2.4.6: Given** that a user with the Operador role attempts to edit a product, **when** executing the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 2.5: As an Administrador, **I want** to deactivate products without deleting them **so that** I can remove them from new quotes and orders while preserving history integrity.

- **Criterion 2.5.1: Given** that the Administrador is in catalog, **when** selecting an active product, **then** the system enables the deactivation option.
- **Criterion 2.5.2: Given** that the Administrador confirms deactivation, **when** the system executes it, **then** it changes status to "Inactivo" and preserves database records.
- **Criterion 2.5.3: Given** that a product is "Inactivo", **when** a virtual assistant prepares a new quote, **then** the system does not offer it among selectable products.
- **Criterion 2.5.4: Given** that a product is "Inactivo", **when** consulting quotes and orders created prior to its deactivation, **then** the product remains visible in those histories.
- **Criterion 2.5.5: Given** that the system manages products, **when** the Administrador reviews available options, **then** no physical product deletion option exists.
- **Criterion 2.5.6: Given** that a user with the Operador role attempts to deactivate a product, **when** executing action, **then** system denies it with the message "No tiene permisos para realizar esta acción".

US 2.6: As an Administrador, **I want** to reactivate previously deactivated products **so that** I can enable them again in new quotes and orders.

- **Criterion 2.6.1: Given** that the Administrador is in catalog, **when** selecting an inactive product, **then** the system enables reactivation option.
- **Criterion 2.6.2: Given** that the Administrador confirms reactivation, **when** the system executes it, **then** it changes status to "Activo".
- **Criterion 2.6.3: Given** that a product was reactivated, **when** a new quote is prepared, **then** the product becomes available for selection again.

US 2.7: As a Potential Customer, **I want** to provide my contact details and required products and quantities to the web portal virtual assistant **so that** I can obtain a quote without contacting an advisor.

- **Criterion 2.7.1: Given** that potential customer accesses virtual assistant on public web portal, **when** initiating interaction, **then** assistant mandatorily requests full name, email address, and phone number before processing request.
- **Criterion 2.7.2: Given** that potential customer omits contact details or enters an email with invalid format, **when** attempting to proceed, **then** assistant does not advance and indicates missing or invalid detail.
- **Criterion 2.7.3: Given** that customer provided contact details, **when** indicating required products and quantities, **then** assistant gathers them from active catalog products.
- **Criterion 2.7.4: Given** that a customer requests an inexistent or deactivated product, **when** the assistant processes the request, **then** it informs it is unavailable and does not generate a price for it.
- **Criterion 2.7.5: Given** that the customer indicates a quantity that is non-numeric, zero, or negative, **when** the assistant processes it, **then** it requests a valid quantity.
- **Criterion 2.7.6: Given** that the customer included products and quantities, **when** requesting a quote, **then** the system automatically calculates an ideal price range for the request estimated dynamically by the AI based on prices used in recent quotes.

US 2.8: As a Potential Customer, **I want** to download my quote as a PDF document directly from the virtual assistant **so that** I can keep the offer details immediately.

- **Criterion 2.8.1: Given** that quote amounts were calculated, **when** the system generates the document, **then** it produces a PDF detailing items, unit prices, totals, and quote validity.
- **Criterion 2.8.2: Given** that the PDF was generated, **when** the assistant delivers it, **then** it offers the customer the option to download it directly.
- **Criterion 2.8.3: Given** that the customer chooses download, **when** confirming the option, **then** the system initiates the PDF download.
- **Criterion 2.8.4: Given** that the download fails, **when** the customer is notified, **then** the assistant allows retrying the download without recalculating or regenerating the quote.

US 2.9: As a Potential Customer, **I want** to receive my quote as a PDF document by email **so that** I can keep the offer details in my inbox and consult them later.

- **Criterion 2.9.1: Given** that the PDF was generated, **when** the assistant delivers it, **then** it offers the customer the option to receive it via email.
- **Criterion 2.9.2: Given** that the customer chooses email delivery, **when** confirming the option, **then** the system immediately sends the PDF as an attachment to the email address provided during the conversation.
- **Criterion 2.9.3: Given** that the email was sent, **when** the process completes, **then** the assistant confirms to the customer that the quote was sent and to which address.

- **Criterion 2.9.4: Given** that the email delivery fails, **when** the system detects the error, **then** the assistant informs the customer and offers to retry or to download the PDF directly.

US 2.10: As an Administrador, **I want** each quote generated by the virtual assistant to be automatically registered in the tracking module **so that** it can be managed subsequently.

- **Criterion 2.10.1: Given** that the virtual assistant generated a quote, **when** the process completes, **then** the system automatically registers it in the internal tracking module.
- **Criterion 2.10.2: Given** that quote was registered, **when** the Administrador consults it, **then** the system displays applicant contact details, quoted products, total amount, and issuance date and time.
- **Criterion 2.10.3: Given** that quote was registered, **when** system assigns initial status, **then** it remains in "Pendiente de revisión" state available for subsequent management.

US 2.11: As an Administrador, **I want** the system to calculate and display a suggested dispatch date on quotes **so that** customers are informed of projected delivery times.

- **Criterion 2.11.1: Given** that the virtual assistant prepares a quote, **when** the customer has defined requested products and volumes, **then** the system calculates an estimated dispatch date and registers it alongside the quote.
- **Criterion 2.11.2: Given** that system calculates suggested date, **when** applying calculation, **then** it uses historical median of real delivery times recorded for requested products and volumes.
- **Criterion 2.11.3: Given** that the Administrador modifies products or quantities of a preliminary quote, **when** saving changes, **then** the system recalculates the suggested date.
- **Criterion 2.11.4: Given** that insufficient historical records exist for requested product or volume combination, **when** system attempts calculation, **then** it displays message "Sin datos históricos suficientes para estimar la fecha de despacho" and suggests no date.
- **Criterion 2.11.5: Given** that quote has a suggested or entered date, **when** the Administrador consults quote management or opens edit interface, **then** date remains visible.

US 2.12: As an Administrador, **I want** to confirm the suggested dispatch date on a preliminary quote **so that** the projected delivery time is formalized as the final dispatch date.

- **Criterion 2.12.1: Given** that a quote in "Pendiente de revisión" or "Modificada" state has a suggested date, **when** the Administrador confirms it, **then** the system keeps it as the final dispatch date.
- **Criterion 2.12.2: Given** that the system could not estimate the date due to missing historical data, **when** the Administrador reviews the quote, **then** the system does not offer the confirmation option because there is no suggested date to confirm.
- **Criterion 2.12.3: Given** that a quote is in "Aprobada" or "Rechazada" state, **when** the Administrador attempts to confirm the dispatch date, **then** the system prevents it because the record is immutable.

US 2.13: As an Administrador, **I want** to adjust the suggested dispatch date on a preliminary quote, or enter it manually when no suggestion exists, **so that** negotiation agreements with customers are accurately reflected.

- **Criterion 2.13.1: Given** that a quote in "Pendiente de revisión" or "Modificada" state has a suggested date, **when** the Administrador adjusts it manually, **then** the system registers the entered date as the final dispatch date.
- **Criterion 2.13.2: Given** that the system could not estimate the date due to missing historical data, **when** the Administrador attempts to approve a quote without entering a date, **then** the system requires manual date entry before proceeding.
- **Criterion 2.13.3: Given** that the Administrador enters a dispatch date earlier than the current date, **when** attempting to save, **then** the system blocks the action and indicates the date is invalid.
- **Criterion 2.13.4: Given** that the Administrador adjusts or enters the date manually, **when** saving, **then** the system logs the change in the audit log alongside details of other quote modifications.
- **Criterion 2.13.5: Given** that a quote is in "Aprobada" or "Rechazada" state, **when** the Administrador attempts to change the dispatch date, **then** the system prevents it because the record is immutable.

US 2.14: As an Operador, **I want** to consult a paginated list of customers **so that** I know registered customers and access their information.

- **Criterion 2.14.1: Given** that the user accesses customer management, **when** the interface loads, **then** the system displays a paginated list of all registered customers.
- **Criterion 2.14.2: Given** that list is visible, **when** the user reviews each record, **then** the system displays business name or full name, NIT or document number, main commercial contact, and status (Activo or Inactivo).
- **Criterion 2.14.3: Given** that list is visible, **when** a user selects a direct access link in a row, **then** the system leads to a unified profile sheet of that customer.
- **Criterion 2.14.4: Given** that list is visible, **when** the user reviews interface options, **then** the system displays actions to register customers, modify customers, and deactivate customers.
- **Criterion 2.14.5: Given** that no registered customers exist, **when** the user accesses the list, **then** the system displays a message indicating no customers available.

US 2.15: As an Operador, **I want** to filter and search customers **so that** I can quickly find customers I need to manage.

- **Criterion 2.15.1: Given** that the customer list is visible, **when** the user filters by NIT or document number, **then** the system displays only matching customers.
- **Criterion 2.15.2: Given** that the customer list is visible, **when** the user filters by business name or full name, **then** the system displays only matching customers.
- **Criterion 2.15.3: Given** that the customer list is visible, **when** the user filters by commercial contact or status, **then** the system displays only customers meeting criterion.
- **Criterion 2.15.4: Given** that the user applies multiple filters simultaneously, **when** executing search, **then** the system displays only customers meeting all criteria.

- **Criterion 2.15.5: Given** that the user searches NIT or documents not corresponding to any registered customer, **when** executing search, **then** the system displays a message indicating the customer does not exist.

US 2.16: As an Operador, **I want** to register new customers **so that** I can associate them with quotes, orders, and inquiries.

- **Criterion 2.16.1: Given** that the user is in customer management, **when** selecting a customer, **then** the system displays a form requiring selection of customer type (natural or legal person).
- **Criterion 2.16.2: Given** that user selected customer type, **when** filling form, **then** the system mandatorily requires business name or full name, NIT or document number, tax or home address, primary phone number, and email address.
- **Criterion 2.16.3: Given** that the user omits a mandatory field, **when** attempting to save, **then** the system blocks action and indicates missing fields.
- **Criterion 2.16.4: Given** that the user enters email with invalid format, **when** attempting to save, **then** the system blocks action and indicates invalid email format.
- **Criterion 2.16.5: Given** that the user enters NIT or document number that already exists, **when** attempting to save, **then** the system prevents creation and displays an alert indicating the document is already registered.
- **Criterion 2.16.6: Given** that data is valid and document is unique, **when** a user saves, **then** the system creates customers in "Activo" state making them available for quotes, orders, and inquiries.

US 2.17: As an Operador, **I want** to modify customer information **so that** I can keep contact details up to date.

- **Criterion 2.17.1: Given** that the user is in customer management, **when** selecting modify a customer, **then** the system displays the form with current data.
- **Criterion 2.17.2: Given** that edit form is open, **when** a user reviews it, **then** it permits modifying business name or full name, address, primary phone number, and email address, displaying person type and NIT or document as non-editable fields.
- **Criterion 2.17.3: Given** an attempt to alter a person type or NIT/document of an existing customer, **when** the system processes a request, **then** it rejects it and preserves original values.
- **Criterion 2.17.4: Given** that a user leaves a mandatory field empty or enters an invalid email, **when** attempting to save, **then** the system blocks action and indicates error.
- **Criterion 2.17.5: Given** that the user modifies data with valid info, **when** saving changes, **then** the system updates information and reflects it immediately in the customer sheet and new registered orders.
- **Criterion 2.17.6: Given** that a customer has previously registered work orders or commercial orders, **when** data is modified, **then** those orders preserve customer data as recorded.

US 2.18: As an Operador, **I want** to deactivate customers without deleting them **so that** I can prevent their use in new orders while preserving historical traceability.

- **Criterion 2.18.1: Given** that the user is in customer management, **when** selecting an active customer, **then** the system enables the deactivation option.
- **Criterion 2.18.2: Given** that the user confirms deactivation, **when** the system executes it, **then** it changes status to "Inactivo" and preserves record and associated data in the database.
- **Criterion 2.18.3: Given** that a customer is "Inactivo", **when** a user registers a new work order, **then** the system does not offer a customer among selectable clients and prevents association with order.
- **Criterion 2.18.4: Given** that a customer is "Inactivo", **when** the user consults the profile and history of quotes and work orders, **then** the system displays them in read-only mode.
- **Criterion 2.18.5: Given** that the system manages customers, **when** the user reviews available options, **then** no physical customer deletion option exists.

US 2.19: As an Operador, **I want** to consult a unified sheet of a customer, identified by NIT or document number, **so that** I can know general and contact information before managing a negotiation.

- **Criterion 2.19.1: Given** that the user is in customer management, **when** accessing the customer sheet from the list, **then** the system displays a sheet identified by NIT or document number.
- **Criterion 2.19.2: Given** that the customer sheet is open, **when** the user consults it, **then** the system displays business name or full name, NIT or document number, address, contact phone numbers, and contact emails.
- **Criterion 2.19.3: Given** that the customer sheet is open, **when** the user consults it, **then** the system consolidates general information, quote and work order history, and commercial notes in a single view.
- **Criterion 2.19.4: Given** that the user accesses NIT or a document not corresponding to any registered customer, **when** the system processes a request, **then** it displays a message indicating the customer does not exist.

US 2.20: As an Operador, **I want** to consult the chronological history of quotes and work orders for a customer **so that** I can evaluate their commercial relationship with the company.

- **Criterion 2.20.1: Given** that the customer sheet is open, **when** the user consults history, **then** the system displays customer quotes and work orders in chronological order.
- **Criterion 2.20.2: Given** that the customer has many records, **when** the user consults history, **then** the system presents them in a paginated format.
- **Criterion 2.20.3: Given** that the customer has no quotes or work orders, **when** the user consults history, **then** the system displays a message indicating no records exist.

US 2.21: As an Operador, **I want** to consult internal commercial notes on a customer's profile sheet **so that** I can review the history and tracking of the relationship with them.

- **Criterion 2.21.1: Given** that the customer sheet is open, **when** the user accesses the notes section, **then** the system displays the registered internal commercial notes in chronological order.

- **Criterion 2.21.2: Given** that the notes section is visible, **when** the user reviews each note, **then** the system displays its content, the registration date and time, and the user who recorded it.
- **Criterion 2.21.3: Given** that a note was modified, **when** the user reviews it, **then** the system identifies it as edited and indicates the date, time, and user of the last modification.
- **Criterion 2.21.4: Given** that the customer has no registered notes, **when** the user consults the section, **then** the system displays a message indicating that no notes exist.
- **Criterion 2.21.5: Given** that notes are for internal use only, **when** the system generates documents or emails addressed to the customer, **then** it does not include the content of the notes.

US 2.22: As an Operador, **I want** to register an internal commercial note on a customer's profile sheet **so that** I can leave a record of the ongoing relationship with them.

- **Criterion 2.22.1: Given** that an active customer's sheet is open, **when** the user selects add a note, **then** the system displays a text field to enter its content.
- **Criterion 2.22.2: Given** that the user attempts to save an empty note, **when** executing the action, **then** the system blocks saving and indicates that the note cannot be empty.
- **Criterion 2.22.3: Given** that the user wrote a valid note, **when** saving it, **then** the system registers it as an internal note, preserves the date, time, and creator user, and displays it in the notes section.
- **Criterion 2.22.4: Given** that the customer is "Inactivo", **when** the user opens their sheet, **then** the system does not allow registering notes because the profile is displayed in read-only mode.

US 2.23: As an Operador, **I want** to modify an internal commercial note that I recorded **so that** I can correct or expand its content.

- **Criterion 2.23.1: Given** that a registered note exists, **when** the user selects edit, **then** the system displays its current content in editable mode.
- **Criterion 2.23.2: Given** that the user leaves the content empty, **when** attempting to save, **then** the system blocks the action and indicates that the note cannot be empty.
- **Criterion 2.23.3: Given** that the user modifies the note with valid content, **when** saving, **then** the system updates the content, preserves the original date, time, and creator user, and records the modification's date, time, and user.
- **Criterion 2.23.4: Given** that the customer is "Inactivo", **when** the user attempts to modify one of their notes, **then** the system prevents it because the profile is in read-only mode.

US 2.24: As an Operador, **I want** to delete an internal commercial note from a customer's profile sheet **so that** I can remove information that is no longer valid or was recorded by mistake.

- **Criterion 2.24.1: Given** that a note exists on an active customer's sheet, **when** the user selects delete, regardless of who recorded it, **then** the system requests specific confirmation.

- **Criterion 2.24.2: Given** that the user confirms deletion, **when** the system executes it, **then** it removes the note from the customer's notes section.
- **Criterion 2.24.3: Given** that the user cancels confirmation, **when** the system processes cancellation, **then** the note is preserved unchanged.
- **Criterion 2.24.4: Given** that a note was deleted, **when** the system logs the operation, **then** it records in the audit log the date, time, responsible user, and the content of the deleted note.
- **Criterion 2.24.5: Given** that the customer is "Inactivo", **when** the user attempts to delete one of their notes, **then** the system prevents it because the profile is in read-only mode.

US 2.25: As an Administrador, **I want** to consult a paginated list of preliminary quotes generated by the virtual assistant **so that** I can supervise pending requests requiring management.

- **Criterion 2.25.1: Given** that the Administrador accesses quote management, **when** the interface loads, **then** the system displays a paginated list with all preliminary quotes generated by the virtual assistant.
- **Criterion 2.25.2: Given** that the list is visible, **when** the Administrador reviews each record, **then** the system displays the issuance date and time, applicant contact details, quoted products, estimated total amount, and status (Pendiente de revisión, Aprobada, Modificada, or Rechazada).
- **Criterion 2.25.3: Given** that the list is visible, **when** the Administrador reviews actions in each row, **then** the system displays options to approve, modify, and reject.
- **Criterion 2.25.4: Given** that a user with the Operator role attempts to access quote management, **when** executing the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 2.26: As an Administrador, **I want** to filter preliminary quotes **so that** I can quickly locate the one I need to manage.

- **Criterion 2.26.1: Given** that the quote list is visible, **when** the Administrador filters by issuance date and time, **then** the system displays only quotes issued within that range.
- **Criterion 2.26.2: Given** that the quote list is visible, **when** the Administrador filters by applicant contact details, **then** the system displays only matching quotes.
- **Criterion 2.26.3: Given** that the quote list is visible, **when** the Administrador filters by quoted products, **then** the system displays only quotes that include those products.
- **Criterion 2.26.4: Given** that the quote list is visible, **when** the Administrador filters by estimated total amount or status, **then** the system displays only quotes meeting the criterion.
- **Criterion 2.26.5: Given** that the Administrador applies multiple filters simultaneously, **when** executing search, **then** the system displays only quotes meeting all criteria.
- **Criterion 2.26.6: Given** that no quote meets applied filters, **when** search executes, **then** the system displays a message indicating no results were found.

US 2.27: As an Administrador, **I want** to approve a preliminary quote **so that** I can formalize the commercial conditions offered to the applicant.

- **Criterion 2.27.1: Given** that a quote is in "Pendiente de revisión" or "Modificada" status, **when** the Administrador approves it, **then** the system changes its status to "Aprobada".
- **Criterion 2.27.2: Given** that the quote was approved, **when** the system logs the change, **then** it records the date, time, and responsible user in the system audit log.
- **Criterion 2.27.3: Given** that the quote was approved, **when** the system completes approval, **then** it automatically sends an email to the applicant attaching the formal official quote document.
- **Criterion 2.27.4: Given** that the quote is in "Aprobada" status, **when** any user attempts to modify it, approve it again, or reject it, **then** the system prevents it because the record is immutable.
- **Criterion 2.27.5: Given** that a quote is in "Rechazada" status, **when** the Administrador attempts to approve it, **then** the system prevents it.
- **Criterion 2.27.6: Given** that a user with the Operador role attempts to approve a quote, **when** executing the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 2.28: As an Administrador, **I want** to modify a preliminary quote **so that** I can adjust its commercial conditions before approving it.

- **Criterion 2.28.1: Given** that a quote is in "Pendiente de revisión" or "Modificada" status, **when** the Administrador selects modify, **then** the system displays the modification interface.
- **Criterion 2.28.2: Given** that the modification interface is open, **when** the Administrador reviews it, **then** it permits adjusting quantities, adding or removing products, modifying applicable unit prices, and estimated delivery time.
- **Criterion 2.28.3: Given** that the Administrador modifies the unit price of a product within the quote, **when** saving, **then** the change applies only to that quote and serves as a new data point for future AI price estimations.
- **Criterion 2.28.4: Given** that the Administrador adds a product, **when** selecting it, **then** the system offers only active catalog products.
- **Criterion 2.28.5: Given** that the Administrador enters an invalid quantity (zero, negative, or non-numeric) or removes all products, **when** attempting to save, **then** the system blocks the action and indicates the error.
- **Criterion 2.28.6: Given** that the Administrador saves valid changes, **when** the system processes them, **then** it automatically recalculates the estimated total amount and updates or maintains the quote status as "Modificada".
- **Criterion 2.28.7: Given** that modifications were saved, **when** the system logs the change, **then** it records the date, time, responsible user, and detail of modifications in the system audit log.
- **Criterion 2.28.8: Given** that a quote is in "Aprobada" or "Rechazada" status, **when** the Administrador attempts to modify it, **then** the system prevents it.
- **Criterion 2.28.9: Given** that a user with the Operador role attempts to modify a quote, **when** executing the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 2.29: As an Administrador, **I want** to reject a preliminary quote by stating a reason **so that** I can discard requests that do not meet commercial conditions and communicate this to the applicant.

- **Criterion 2.29.1: Given** that a quote is in "Pendiente de revisión" or "Modificada" status, **when** the Administrador selects reject, **then** the system mandatorily requests a reason or justification for rejection.
- **Criterion 2.29.2: Given** that the Administrador does not enter a reason, **when** attempting to confirm rejection, **then** the system blocks the action and indicates that the reason is mandatory.
- **Criterion 2.29.3: Given** that the Administrador enters the reason and confirms, **when** the system executes rejection, **then** it changes status to "Rechazada" and preserves the record.
- **Criterion 2.29.4: Given** that the quote was rejected, **when** the system logs the change, **then** it records the date, time, responsible user, and reason in the system audit log.
- **Criterion 2.29.5: Given** that the quote was rejected, **when** the system completes rejection, **then** it automatically sends an email to the applicant informing them of the rejection and justification.
- **Criterion 2.29.6: Given** that the quote is in "Rechazada" status, **when** any user attempts to modify or approve it, **then** the system prevents it because the record is immutable.
- **Criterion 2.29.7: Given** that a user with the Operador role attempts to reject a quote, **when** executing the action, **then** the system denies it with the message "No tiene permisos para realizar esta acción".

US 2.30: As an Administrador, **I want** approved commercial orders to automatically enter a processing queue **so that** they are available for consolidation and export to the accounting system.

- **Criterion 2.30.1: Given** that a commercial order reaches "Aprobada" status, **when** the system logs approval, **then** it automatically incorporates it into the administrative processing queue.
- **Criterion 2.30.2: Given** that an order enters the queue, **when** the system registers it, **then** it saves date, time, and unique order identifier, leaving it in waiting mode.
- **Criterion 2.30.3: Given** that an order is in preliminary, pending review, modified, or rejected status, **when** the system evaluates the queue, **then** it does not permit entry into said flow.
- **Criterion 2.30.4: Given** that an order is already in queue, **when** the system re-processes approval, **then** it does not generate a duplicate record.

US 2.31: As an Operador, **I want** to consult the queue of approved orders **so that** I can know which orders are available for export to the accounting system.

- **Criterion 2.31.1: Given** that the user accesses the processing queue, **when** the interface loads, **then** the system displays a list of available approved commercial orders.

- **Criterion 2.31.2: Given** that list is visible, **when** the user reviews each record, **then** the system displays approval date and time, customer, products, total amount, and export status.
- **Criterion 2.31.3: Given** that the queue has no orders, **when** the user accesses the interface, **then** the system displays a message indicating no orders available.

US 2.32: As an Operador, **I want** to filter orders in the queue **so that** I can quickly find those needed to build a batch.

- **Criterion 2.32.1: Given** that the queue is visible, **when** the user filters by date range, **then** the system displays only orders approved within that range.
- **Criterion 2.32.2: Given** that the queue is visible, **when** the user filters by the customer, **then** the system displays only orders for that customer.
- **Criterion 2.32.3: Given** that the queue is visible, **when** the user filters by order identifier, **then** the system displays only matching orders.
- **Criterion 2.32.4: Given** that the user applies multiple filters simultaneously, **when** executing search, **then** the system displays only orders meeting all criteria.
- **Criterion 2.32.5: Given** that no order meets filters, **when** search executes, **then** the system displays a message indicating no results were found.

US 2.33: As an Operador, **I want** to select multiple orders from the processing queue **so that** I can build the batch to export to the accounting system.

- **Criterion 2.33.1: Given** that the queue is visible, **when** the user checks multiple orders using checkboxes, **then** the system includes them in the export batch.
- **Criterion 2.33.2: Given** that the user checked orders, **when** unchecking any, **then** the system excludes it from the export batch.
- **Criterion 2.33.3: Given** that an order was already exported, **when** the user reviews the queue, **then** the system does not allow including it in a new batch.

US 2.34: As an Operador, **I want** to export selected orders into a single flat file for Siigo Pyme **so that** manual re-entry of information into the accounting system is avoided.

- **Criterion 2.34.1: Given** that the user selected no orders, **when** attempting to export, **then** the system blocks action and indicates at least one order must be selected.
- **Criterion 2.34.2: Given** that the user selected orders, **when** executing export, **then** the system generates a single flat file formatted strictly according to Siigo Pyme specifications for batch load.
- **Criterion 2.34.3: Given** that file was generated, **when** the process completes, **then** the system permits the user to download it.
- **Criterion 2.34.4: Given** that export executes, **when** the process completes, **then** the system only generates a file without automatically writing to Siigo Pyme or real-time syncing.
- **Criterion 2.34.5: Given** that the file was generated, **when** the system logs operation, **then** it records date, time, responsible user, and batch detail in the system audit log.
- **Criterion 2.34.6: Given** that the file was generated, **when** the system completes the process, **then** it marks orders as exported and excludes them from the active queue to prevent duplication in subsequent downloads.

- **Criterion 2.34.7: Given** that file generation fails, **when** the system detects an error, **then** it displays an error message, does not mark orders as exported, and retains them in the active queue.

US 2.35: As an Operador, **I want** to consult a paginated list of work orders **so that** I can control their lifecycle.

- **Criterion 2.35.1: Given** that user accesses work order management, **when** interface loads, **then** system displays a paginated list with all registered O.T.
- **Criterion 2.35.2: Given** that the list is visible, **when** the user reviews each record, **then** the system displays O.T. number, customer info, item details, agreed delivery date, and status (Pendiente, Registrada en Siigo, Conciliada, or Anulada).
- **Criterion 2.35.3: Given** that the list is visible, **when** the user reviews options, **then** the system displays access to register new O.T. and, in each row, edit and annul actions.
- **Criterion 2.35.4: Given** that no registered O.T. exist, **when** the user accesses a list, **then** the system displays a message indicating no work orders exist.

US 2.36: As an Operador, **I want** to filter work orders **so that** I can quickly locate the one I need to manage.

- **Criterion 2.36.1: Given** that O.T. list is visible, **when** the user filters by O.T. number, **then** the system displays only matching orders.
- **Criterion 2.36.2: Given** that O.T. list is visible, **when** the user filters by customer, **then** the system displays only orders for that customer.
- **Criterion 2.36.3: Given** that O.T. list is visible, **when** the user filters by item details, **then** the system displays only orders containing those items.
- **Criterion 2.36.4: Given** that O.T. list is visible, **when** the user filters by agreed date or status, **then** the system displays only orders meeting criterion.
- **Criterion 2.36.5: Given** that the user applies multiple filters simultaneously, **when** executing search, **then** the system displays only orders meeting all criteria.
- **Criterion 2.36.6: Given** that no O.T. meets filters, **when** search executes, **then** the system displays a message indicating no results were found.

US 2.37: As an Operador, **I want** to manually register a new work order **so that** I can manage jobs requested by customers from creation.

- **Criterion 2.37.1: Given** that the user is in O.T. management, **when** selecting to register a new order, **then** the system displays a form requiring customer, item details with quantities, and agreed delivery date.
- **Criterion 2.37.2: Given** that the user selects a customer or items, **when** the system displays options, **then** it offers only active customers and active products.
- **Criterion 2.37.3: Given** that the user omits a mandatory field or adds no items, **when** attempting to save, **then** the system blocks action and indicates what is missing.
- **Criterion 2.37.4: Given** that the user enters an invalid quantity (zero, negative, or non-numeric), **when** attempting to save, **then** the system blocks action and indicates error.

- **Criterion 2.37.5: Given** that the user enters an agreed date earlier than the current date, **when** attempting to save, **then** the system blocks action and indicates the date is invalid.
- **Criterion 2.37.6: Given** that data is valid, **when** the user saves, **then** the system automatically assigns a unique, consecutive O.T. number and sets initial status as "Pendiente".
- **Criterion 2.37.7: Given** that O.T. was created, **when** the system logs operation, **then** it records date, time, and creator in the system audit log.

US 2.38: As an Operador, **I want** to modify an existing work order **so that** I can correct or update its information while its cycle is not concluded.

- **Criterion 2.38.1: Given** that an O.T. is in "Pendiente" or "Registrada en Siigo" status, **when** the user selects edit, **then** the system displays the form with current data.
- **Criterion 2.38.2: Given** that edit form is open, **when** user reviews it, **then** it permits adjusting customer, adding, modifying, or removing items with quantities, and updating agreed date, displaying O.T. number as non-editable.
- **Criterion 2.38.3: Given** an attempt to alter an O.T. number, **when** the system processes a request, **then** it rejects it and preserves the original number.
- **Criterion 2.38.4: Given** that the user leaves a mandatory field empty, removes all items, or enters an invalid quantity or date, **when** attempting to save, **then** the system blocks action and indicates error.
- **Criterion 2.38.5: Given** that the user modifies data with valid info, **when** saving changes, **then** the system updates the record.
- **Criterion 2.38.6: Given** that modifications were saved, **when** the system logs change, **then** it records date, time, responsible user, and detail of modified data in the system audit log.
- **Criterion 2.38.7: Given** that an O.T. is in "Conciliada" or "Anulada" status, **when** a user attempts to edit it, **then** the system prevents it.

US 2.39: As an Operador, **I want** to annul a work order stating a reason **so that** I can render void orders that should not continue while preserving record.

- **Criterion 2.39.1: Given** that an O.T. is in "Pendiente" or "Registrada en Siigo" status, **when** user selects annul, **then** system mandatorily requests a reason or justification for annulment.
- **Criterion 2.39.2: Given** that the user does not enter reason, **when** attempting to confirm annulment, **then** the system blocks action and indicates reason is mandatory.
- **Criterion 2.39.3: Given** that the user enters reason and confirms, **when** the system executes annulment, **then** it changes status to "Anulada" and preserves record in history.
- **Criterion 2.39.4: Given** that O.T. was annulled, **when** the system logs operation, **then** it records date, time, responsible user, and reason in system audit log.
- **Criterion 2.39.5: Given** that an O.T. is in "Conciliada" or already "Anulada" status, **when** the user attempts to annul it, **then** the system prevents it.

US 2.40: As an Operador, **I want** to update work order status following its established workflow **so that** its progress up to accounting integration and reconciliation is reflected.

- **Criterion 2.40.1: Given** that an O.T. is in "Pendiente" status, **when** the user updates status, **then** the system permits advancing only to "Registrada en Siigo" or annulling it.
- **Criterion 2.40.2: Given** that an O.T. is in "Registrada en Siigo" status, **when** the user updates status, **then** the system permits advancing only to "Conciliada" or annulling it.
- **Criterion 2.40.3: Given** that the user attempts a status change between non-consecutive statuses (e.g., from "Pendiente" to "Conciliada"), **when** executing the action, **then** the system prevents it and indicates allowed transition.
- **Criterion 2.40.4: Given** that an O.T. is in "Conciliada" or "Anulada" status, **when** a user attempts a status change, **then** the system prevents it.
- **Criterion 2.40.5: Given** that an O.T. status is updated, **when** the system logs change, **then** it records date, time, and responsible user in the system audit log.

User Story Map

Due to the extensive size and level of detail of the user story map, we recommend consulting it in its entirety directly through the following hyperlink.



Declaration of the Use of Artificial Intelligence

For the preparation of this document, we declare that the use of Artificial Intelligence tools -specifically Gemini and Claude- was strictly limited to support, review, and translation purposes. First, these tools were used exclusively to translate the content into English. Likewise, we relied on them to conduct brainstorming sessions to explore new proposals, a process that was carried out only after we had independently defined and structured the system's core functionalities.

Furthermore, Gemini and Claude were employed as a validation mechanism to ensure there were no weak points or gaps in the project's overall reasoning. This included reviewing the definition of requirements, user stories, and the user story map to guarantee their logical soundness. Finally, we used both artificial intelligences to confirm that each of these elements strictly complied with the correct structure and format required for the submission.